RIET-P

a chamber organ for the M-VAVE FM-1

The Player's Manual

Four wooden stops · forty-eight pipes speaking at once · the wind,
the plate and the old temperaments

The Instrument

KIST-P is KIST with its rooms taken out and a plate put in: a large sheet of steel hung in a frame, set ringing by a driver and heard through two pickups, as studios have used since the 1950s for an echo that can be made to measure. Every part of it is yours to set, on three pages of its own. The organ itself is the same as KIST's in every way.

KIST turns the FM-1 into a small wooden pipe organ of the kind that sits beside a harpsichord in a baroque ensemble: a box organ, or kistorgel as the Dutch call it, built in the manner of the continuo organs of the Klop workshop. It has one keyboard, four stops and nothing else, and it is meant to be played rather than programmed.

Nothing you hear is a recording. An ordinary digital organ plays back samples: recordings of real pipes, the same recording every time a key goes down. KIST does something different. It works out, many thousands of times a second, what the air is doing inside every pipe that is speaking: the jet of wind at the mouth, the column of air in the pipe, the way the two push each other into a steady tone. The sound is the result of that calculation, made fresh for every note.

That has consequences you can hear and feel:

- Every note starts a little differently. A pipe takes a moment to settle, with a small breathy 'chiff', and how it settles depends on how quickly you press the key.

- The pipes share their wind. Hold a big chord and the wind sags a touch, as it does in a real organ whose bellows are working hard; switch the blower off and the notes die as the reservoir empties.
- No two pipes are quite alike. Each is tuned a hair's breadth away from perfect, as real pipes are, so a chord has life in it rather than the glassy stillness of a machine.
- The tuning is truly the old tuning. When you choose meantone or Werckmeister, the pipes themselves are tuned to it; you are not hearing a modern organ bent out of shape.

The keyboard runs from the C two octaves below middle C to the F above the C two octaves above it: 54 keys, the compass of a small continuo organ. The FM-1's own keys play the middle of it; a larger keyboard plugged in over MIDI reaches the rest (see *Playing from another keyboard*).

The four stops

A stop is a rank of pipes, one for each key. Pulling a stop out lets the wind into its rank; on KIST, pressing its button does the same, and the button lights while the stop is drawn.

Stop	Pitch	Its sound
Gedackt	8 foot	Stopped wooden pipes, capped at the top: soft, round and flute-like, sounding at the pitch you play. The foundation of everything; on its own, the classic continuo sound.
Rohrflöte	4 foot	A chimney flute: a stopped pipe with a little open tube in its cap, which gives it a brighter, slightly nasal edge. It sounds an octave higher than you play.
Principal	2 foot	Open wooden pipes, two octaves up: clear and singing. It adds brilliance and makes the organ carry.
Quinta	2⅔ foot	Open pipes sounding a twelfth higher (an octave and a fifth). Only in the treble. It is a colour stop: on its own it sounds odd, with the Gedackt it blends into a reedy, singing solo voice.

"Foot" is the organ builder's old way of naming pitch: an 8-foot pipe is the length that sounds the note you play at the bottom of the keyboard; a 4-foot pipe, half as long, sounds an octave higher; and so on.

Bass and treble

Every stop is divided between the bass half of the keyboard and the treble half, and each half has its own button. This is how the small Dutch organs were built, and it is very useful: you can accompany softly in the left hand on the Gedackt alone while the

right hand plays a brighter solo with the Principal or the Quinta added. The division falls between B and middle C, and can be moved (see *The Works*).

Some registrations to start with

For	Draw
Continuo, accompanying voices or strings	Gedackt 8 alone, both halves
A little more presence	Gedackt 8 and Rohrflöte 4
Full organ for a chorale	Gedackt 8, Rohrflöte 4, Principal 2
A solo in the right hand	Gedackt 8 and Quinta in the treble, Gedackt 8 alone in the bass
A bright, silvery solo	Gedackt 8 and Principal 2 in the treble

Finding your Way

All of KIST-P is played from the FM-1's own buttons and knobs. The words printed on the panel belong to the FM-1's original firmware; under KIST-P they do these things:

The buttons

Button	In KIST-P
FX / SEL	Gedackt 8, bass / treble
ENV / LFO	Rohrflöte 4, bass / treble
EDIT / GLO	Principal 2, bass / treble
ARP	Quinta 2⅔ (treble only)
REC	Cancel: put every stop in at once. Press again to bring them all back.
SEQ	The tremulant: a gentle, regular wavering of the wind
PLAY	The blower. Green while it runs.
SAVE	Keep the present setup for the next time the FM-1 is switched on. Lit while something has changed and is not yet kept.
HOME	Back to the first page, the Console
OCT- / OCT+	Move the FM-1's own keys down or up an octave; both together put them back

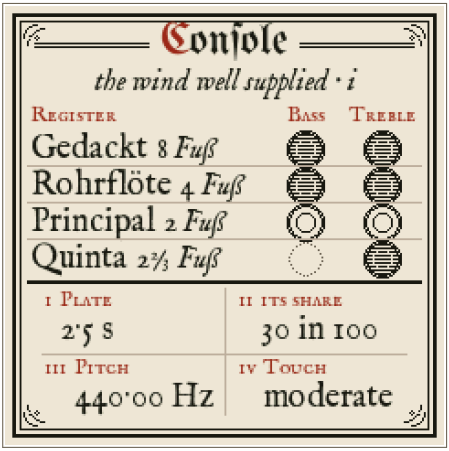
The knobs

Knob	In KIST-P
SELECT	Turn the pages of the screen
PRESETS	How much of the plate you hear (the mix), from any page
ALGORITHM	Choose the temperament, from any page
MASTER	The volume
The four knobs K1–K4	Whatever the page shows at its foot, in the same arrangement as the knobs themselves

Reading the screen

Every page is laid out the same way, like a page from an old printed book. At the top is its title in black-letter, and under it a line in italics that says what the page is about. When you turn a knob, that line briefly shows, in red, what you have just changed.

The middle of the page is its picture: the stops, the tuning, the plate, the wind. At the foot is a small table of four boxes, numbered i to iv. They stand for the four knobs, in the same places: i is the first knob, ii the second, and so on. Each box has the knob's name in small red letters and, under it, what it is set to now.



The Console page. Each round head is a stop half: filled when drawn. The Quinta has no bass half, so its left head is dotted.

The Pages

Turn SELECT *to move through the eight pages.* HOME *always brings you back to the first.*

I · The Console

The organ's front: the four stops with their bass and treble halves, and the four things you are most likely to want while playing.

Knob	Sets
i · Decay	How long the plate rings, in seconds (see *The Plate*)
ii · its share	How much of the plate you hear, from 0 (the pipes alone) to 100
iii · Pitch	The pitch of A (see *Tuning*)
iv · Touch	How the keys respond, from *soft* to *crisp* (see *The Wind*)

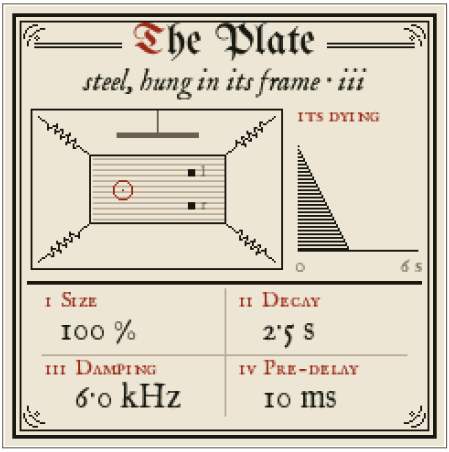
II · Tuning

Where the organ is tuned, and how. This page has its own chapter, *Tuning*, below.

III · The Plate

An organ is never heard on its own: the building is half the instrument. KIST-P gives it a plate instead: the picture is the plate in its frame, hung on four springs, with the driver (the red

ring) that sets it moving, the two pickups that hear it, and a felt damper that closes on it as you take the treble out. The plate grows and shrinks with its size. Under it, the curve is how the sound dies away over six seconds.

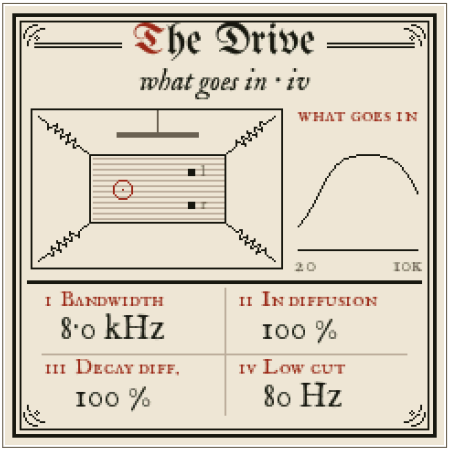


The Plate. The drawing follows the size and the damping; the curve below is the decay.

Knob	Sets
i · Size	How large the plate is, 30 to 130 %. A larger plate is darker and its echoes are further apart. The size glides rather than jumps, so turning it while the plate rings bends the pitch of the tail, a sound no real plate can make.
ii · Decay	How long the plate rings, 0.2 to 12 seconds. It holds while the size glides.
iii · Damping	Where the treble starts to die sooner than the bass, 1 to 10 kHz: low is a dark, felted plate, high a bright, bare one
iv · Pre-delay	A pause before the plate answers, 0 to 100 ms: a little keeps the pipes' attack clear of the echo

IV · The Drive

What goes into the plate, and how it spreads. The curve is the plate's colour, bass on the left, treble on the right.

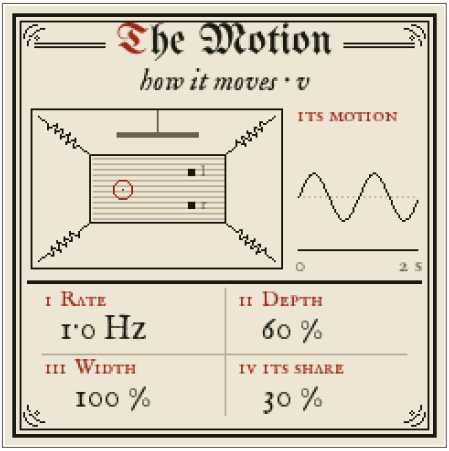


The Drive. The curve shows the bandwidth going in and the low cut.

Knob	Sets
i · Bandwidth	How much treble reaches the plate, 1 to 10 kHz
ii · Input diffusion	How quickly a note is smeared into a wash as it enters: low keeps some separate early echoes
iii · Decay diffusion	How dense the tail is: low, it grows grainy and you hear its echoes
iv · Low cut	Takes the deep bass out of the plate, off to 500 Hz, so the Gedackt's low notes don't boom

v · The Motion

A real plate is never quite still, and a little movement keeps a long tail from ringing on one note. The sine shows how fast and how far it moves.



The Motion: the rate and depth of the plate's wander, its width and the mix.

Knob	Sets
i · Rate	How fast the plate wanders, 0.1 to 3 times a second
ii · Depth	How far, from still (0) to 100. Deep and fast makes a chorus of it.
iii · Width	From one point in the middle (0) to the full width of the two pickups (100)
iv · Mix	How much of the plate you hear, as PRESETS sets it

VI · The Wind

The wind is what makes an organ breathe. KIST models a reservoir of air, filled by the blower and drawn on by the pipes.

The page shows the reservoir as a pair of bellows and a number (100 is the pressure the pipes were voiced for), and under it a trace of the last six seconds, so you can watch the wind dip when you play a full chord, or rock with the tremulant.

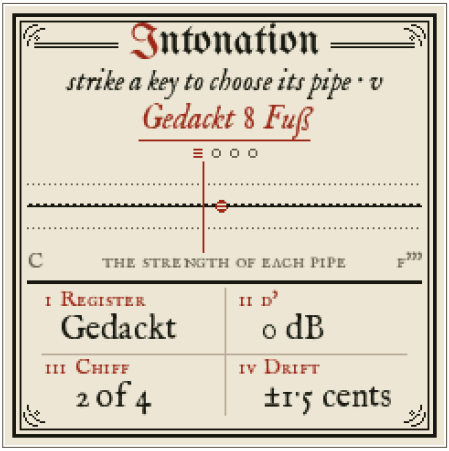


The Wind, with the tremulant beating. The line under the title tells you how hard the last key was struck and how quickly its pipe valve opened.

Knob	Sets
i · Wind	The pressure, a little above or below the voiced pressure. Less wind makes the pipes softer and a touch flatter, more makes them firmer and sharper, as on a real organ.
ii · Steadiness	*steady*, *lively* or *loose*: how much the wind gives way when many pipes speak at once
iii · Tremulant	How fast the tremulant beats, in beats a second
iv · its depth	*gentle*, *moderate* or *deep*

VII · Intonation

Intonation is what an organ builder does after the pipes are made: going through them one by one, making each sit well with its neighbours. KIST lets you do the same. Choose a stop with the first knob, then strike a key: that pipe is marked in red on the chart, and the second knob makes it louder or softer.

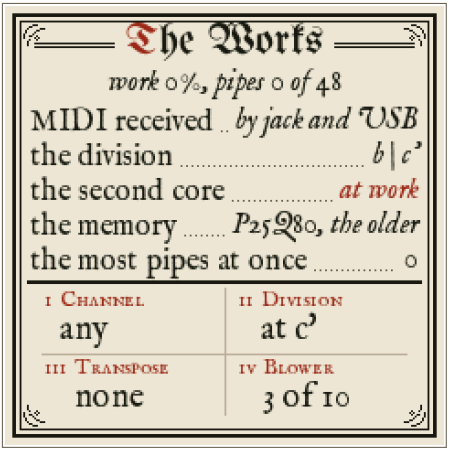


Intonation. Each mark is one pipe of the chosen stop; above the line is louder, below softer. The red upright line is the division between bass and treble.

Knob	Sets
i · Register	Which stop you are working on
ii · (the note)	How loud the last pipe you struck is, in half-decibel steps
iii · Chiff	How much of a breathy start the stop's pipes have, from none to 4
iv · Drift	How far each pipe is allowed to wander from perfect tuning, in cents (hundredths of a semitone). A little drift makes chords sound alive; none makes them sound electronic.

viii · The Works

The organ's workings: things you set once and leave.



The Works. The line under the title shows how hard the FM-1 is working, and how many pipes are speaking.

Knob	Sets
i · Channel	Which MIDI channel KIST listens to, or *any*
ii · Division	Where the bass ends and the treble begins
iii · Transpose	Shift everything you play over MIDI up or down by semitones, for music written at a different pitch
iv · Blower	The faint hum and rush of the organ's blower, from silent to 10 (on KIST it is on the Room page)

The page also says which memory chip your FM-1 has (there are two kinds, and KIST knows both), and whether the FM-1's second processor is helping. Most FM-1s have a second processor that KIST uses for the highest pipes and the plate. On some units it is unreliable; KIST notices, stops using it and carries on with one,

and the page then says retired. You will not hear the difference in ordinary playing.

Tuning

Two questions decide how an organ is tuned: how high it is pitched, and how the twelve notes of the octave are shared out between them. KIST lets you choose both, the way organs were tuned when this music was new.

The pitch of A

Today an orchestra tunes its A to 440 vibrations a second. It was not always so. Baroque ensembles commonly play at 415, a semitone lower; French baroque music often at 392, a tone lower; some Venetian music at 465. KIST can be set anywhere from 300 to 500.99. Turning the first knob on the Tuning page moves in whole numbers (a quick turn jumps by ten) and pauses for a moment at 392, 415, 432, 440 and 465, so the usual pitches are easy to find. The second knob sets the hundredths, for matching another instrument exactly.

Temperament

There is no way to tune twelve notes so that every interval in every key is perfectly pure: the arithmetic does not close. Something has to give, and a *temperament* is a decision about where. Modern equal temperament spreads the error evenly, so every key sounds the same and none sounds quite sweet. The older temperaments made the common keys beautiful and let the rare ones grow sharp and colourful. Music written for them sounds as its composers heard it when played in them.

Temperament	What it is like
Equal	Every key alike: the modern piano. Safe for anything, sweet in nothing.
Werckmeister III	1691. Pure, bright keys near C; the far keys more restless. Good for Bach and Buxtehude.
Meantone	The Renaissance and early baroque sound: the most beautiful thirds, but only a few keys usable. One fifth, the 'wolf', howls; music in the right keys never goes near it.
Kirnberger III	1779. Very pure in the common keys, still playable in all of them.
Vallotti	About 1750. Gentle and well balanced: nearly equal, but each key keeps a little of its own character. A good everyday choice.

Turn ALGORITHM from any page to choose. A list appears; KIST waits until the knob has been still for a moment before retuning, so you can turn past several without the organ jumping about.

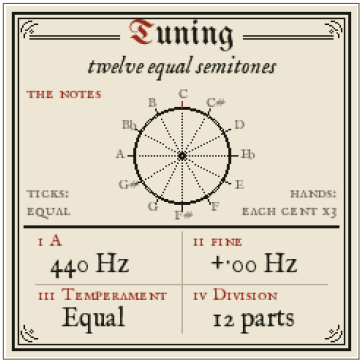


Choosing a temperament with ALGORITHM.

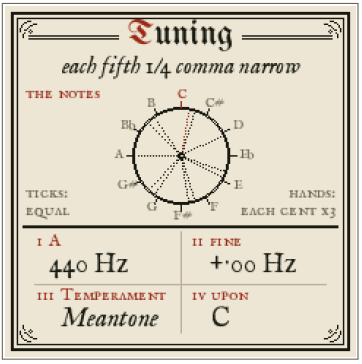
Every temperament can also be moved to start on another note with the fourth knob (*upon*), as tuners did to favour the keys a piece is written in.

Reading the clock

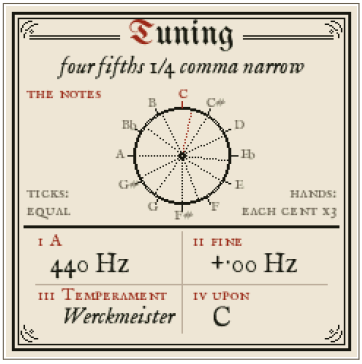
The Tuning page draws the octave as a clock face, with C at twelve o'clock and the other eleven notes round the dial in order. The short ticks outside the ring are where equal temperament puts each note. The dotted hands are where the temperament you have chosen puts them. Where a hand points straight at its tick, that note is just as in equal temperament; where it leans away, the note is higher (clockwise) or lower. The differences are small, so the clock draws them three times larger than they are.



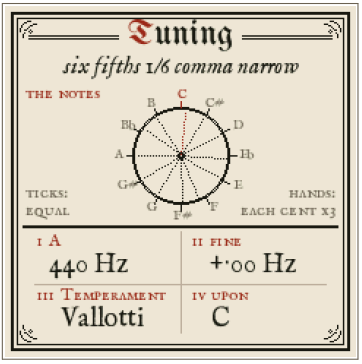
Equal: every hand on its tick.



Meantone: the hands lean far, G# and Eb most.



Werckmeister III.

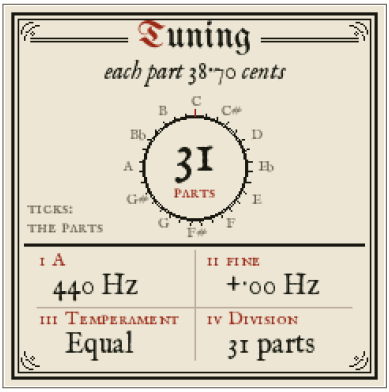


Vallotti: smaller leans, mostly in the sharps and flats.

The red hand is the note the temperament is laid out from. Move it with *upon* and the whole pattern turns with it.

Equal divisions

Choose *Equal* and the fourth knob sets how many equal steps the octave is divided into, from 5 to 72; 12 is the ordinary keyboard. Other numbers are a door to other musical worlds. With 19 or 31 steps the thirds come close to meantone's sweetness while every key works; 31 was proposed for this in the seventeenth century, and an organ tuned so stood in Haarlem. With other divisions, each key of the keyboard plays the next step, so the octave no longer falls on the same letter. The clock then shows the steps as ticks round the dial.



Thirty-one equal parts. Every key plays the next of the thirty-one steps.

Touch and Wind

On a mechanical organ the key is joined to the pipe valve by a long wooden rod. You cannot make a pipe louder by striking harder, but you can feel and hear how quickly the valve opens: slowly, and the pipe speaks gently; quickly, and it speaks with a little more attack.

KIST works the same way. How hard you strike a key changes how fast its valve opens, not how loud the pipe is. The fourth knob on the Console sets how strongly your touch matters, from *soft* (every note starts gently) to *crisp* (a firm stroke gives a quick, clear start).

Things worth trying:

- Hold a full chord with every stop drawn and watch the Wind page: the reservoir drops a little, and the pitch with it, as on a real organ. Set *Steadiness* to *loose* to exaggerate it.
- Switch the blower off with PLAY while holding a chord. The notes go on as long as there is wind in the reservoir, then sink and die.
- Press SEQ for the tremulant. It is a proper wind tremulant: it shakes the air, so the pitch and loudness waver together.
- Draw or retire a stop while holding keys. The pipes come in and out as they would on a real slider chest.
- Press REC to silence every stop at once (to change registration quickly between pieces), then press it again to bring them all back.

Saving

When you have the organ as you like it, press SAVE. The next time the FM-1 is switched on, it will come back exactly so: the stops, the plate, the tuning, the wind and your voicing. Saving takes a moment during which the organ falls silent, so save between pieces rather than during one. While something has changed and not been saved, the SAVE button is lit.

Playing from another keyboard

The FM-1's own keyboard is small. A larger keyboard plugged in by USB or into the FM-1's MIDI socket plays the whole 54-key compass. The sustain pedal does nothing, as on an organ: a pipe sounds exactly as long as its key is held. Velocity from the keyboard works as touch, as above.

Installing and Going Back

KIST replaces the FM-1's own firmware, and can always be replaced by it again.

The easiest way is *FM-1 Manager*, a small program for Mac, Windows and Linux, from the KIST page on GitHub (github.com/78Hz/kist, under *Releases*):

- Before the first install, use FM-1 Manager's *Whole flash* tab to keep a complete copy of the FM-1 as it came from the factory (Mac and Linux).
- Connect the FM-1 to the computer with a USB cable, switched on as usual.
- In the *Firmware* tab, choose KIST-P and press *Install on FM-1*. It takes a few minutes; do not unplug it until it says it has finished.
- To go back, choose M-VAVE's own firmware the same way (the Manager can fetch it for you).

If anything goes wrong during an install, install again: the FM-1 keeps its old firmware until the new one is safely written. The Manager's *Saved data* tab keeps a copy of your saved setup, so a new version of KIST, or a different FM-1, can have your organ back.

Credits

KIST is free software under the Mozilla Public License 2.0, built on fm1-base and felucca-bsp. Its screens and this manual are set in IM Fell English, cut by Igino Marini after the types Bishop John Fell gave to Oxford in the seventeenth

century, and UnifrakturMaguntia, both under the SIL Open Font License. KIST is inspired by the box organs of the Dutch Klop workshop and is not connected with it. M-VAVE and FM-1 are trademarks of their owners; KIST is not affiliated with M-VAVE.

At a Glance

Page	I	II	III	IV
Console	Decay	its share	Pitch	Touch
Tuning	A (Hz)	fine	Temperament	upon / Division
The Plate	Size	Decay	Damping	Pre-delay
The Drive	Bandwidth	In. diffusion	Dec. diffusion	Low cut
The Motion	Rate	Depth	Width	Mix
The Wind	Wind	Steadiness	Tremulant	its depth
Intonation	Register	the note	Chiff	Drift
The Works	Channel	Division	Transpose	Blower

Control	Does
FX SEL ENV LFO EDIT GLO ARP	The stops, bass and treble
REC	Cancel all stops /bring them back
SEQ⁀	Tremulant
PLAY	Blower
SAVE	Keep this setup
HOME	The Console
SELECT	Turn the pages
PRESETS	Plate mix
ALGORITHM	Temperament
MASTER	Volume